LeakTrace on AWS: Architecture

Region ap-south-1 (Mumbai) · every printed exam copy carries a hidden fingerprint, so a leaked paper can be traced to its exact seat

Users (web browser)
One login page, three roles

- **Controller** · writes template, generates copies, schedules, traces leaks
- **Superintendent (SI)** · downloads own centre's papers inside the window
- **Auditor** · read-only audit log, alerts, trace cases

`http://ec2-13-233-233-79.ap-south-1.compute.amazonaws.com`

Amazon CloudFront
Planned, not active

Blocked until AWS verifies the account. nginx on the API server stands in for it (one address for the site and /api).

Developer laptop
VS Code + Claude Code

npm run package builds:

api.tar.gz · worker.tar.gz · web.tar.gz

Uploaded to the **S3 deploy bucket** in the console. Same code runs locally or on AWS: `APP_MODE=aws` swaps adapters (S3, SQS, SNS, Scheduler, Parameter Store, RDS).

Email inboxes
SNS subscribers

Leak alerts, ops alerts and release notices arrive here (2 people per topic).

AWS Cloud Region ap-south-1 (Mumbai) · account 0840-3233-3359

VPC **leaktrace-vpc** · 10.0.0.0/16

Availability Zone ap-south-1a

Public subnet 10.0.1.0/24

Security group **api-sg** · inbound HTTP 80 from allowed IP

IGW Internet Gateway

HTTP :80

Amazon EC2 · leaktrace-api
t3.micro · Amazon Linux 2023 · public IP

- **nginx :80** · serves React site from /var/www/leaktrace, proxies `/api`
- **Node.js API :3000** · Express, systemd service, APP_MODE=aws
- **Exam-events consumer** · reads RELEASE / CLOSE messages

IAM role: leaktrace-api-role

SQL :5432 (TLS)

Private subnet 10.0.11.0/24 · no internet route

Security group **db-sg** · 5432 only from api-sg + worker-sg

Amazon RDS · PostgreSQL 16
leaktrace-db · db.t4g.micro · single-AZ

`users` `centres · rooms` `exams` `paper_templates` `copies (fingerprints)` `generation_jobs` `worker_heartbeats` `downloads` `trace_cases` `audit_events (append-only)`

TLS connections · encrypted at rest · app user leaktrace_app

Availability Zone ap-south-1b

boot: servers download code

Public subnet 10.0.2.0/24

Security group **worker-sg** · no inbound rules

EC2 Auto Scaling group · worker
min 0 · max 1 · launch template

- **Worker instance** (t3.micro) · `node dist/main.js`
- Renders each copy as PDF (pdfkit), merges one PDF per room
- Boot script downloads **worker.tar.gz** from S3; heartbeat every 15 s

IAM role: leaktrace-worker-role · costs $0 when idle

SendMessage: one job per room

Private subnet 10.0.12.0/24 · no internet route

exam-events → API marks RELEASED / CLOSED

DB subnet group
RDS requires subnets in two AZs. The database runs in 1a; this subnet keeps 1b ready for a standby or failover.

Route tables
Public: 10.0.0.0/16 local · 0.0.0.0/0 → Internet Gateway · S3 → endpoint
Private: local only + S3 → endpoint · **no NAT gateway** (saves cost)

progress + heartbeat (SQL)

CreateSchedule (release / close)

VPCE S3 gateway endpoint

PutObject PDFs private path

Amazon S3
3 buckets

- **leaktrace-deploy-ccl2026** · api / worker / web .tar.gz; servers download at boot
- **leaktrace-web-ccl2026** · React build (source copy for nginx / future CloudFront)
- **leaktrace-data-ccl2026** · room PDFs + leak evidence · lifecycle: delete after 7 days · pre-signed links 120 s

Amazon SNS
3 topics → email

- **leaktrace-leak-alerts** · a leak was traced
- **leaktrace-ops-alerts** · system problems, DLQ
- **leaktrace-release-notices** · paper released

Publish alerts / release notices

Amazon CloudWatch
metrics + alarms

- **queue-has-work** · messages ≥ 1 → scale worker to 1
- **queue-empty** · 0 for 5 min → scale worker to 0
- **dlq-not-empty** · → ops-alerts email

depth

scale out / in

Amazon SQS
3 queues

- **leaktrace-generation** · one job per room; long poll 20 s
- **leaktrace-generation-dlq** · dead-letter queue after 3 failed receives
- **exam-events** · RELEASE / CLOSE events from the scheduler

ReceiveMessage

fires at release time

EventBridge Scheduler
schedule group **leaktrace**

One-time `at()` schedules for release and close, India time. Target: exam-events queue via **leaktrace-scheduler-role**.

AWS Systems Manager
config + access

- **Parameter Store** /leaktrace/prod · 16 settings, secrets as SecureString
- **Session Manager** · server terminal from the console, no SSH / port 22

settings read at start-up

AWS IAM
least-privilege roles, no keys in code

- **api-role** · SSM, S3, SQS, SNS, Scheduler
- **worker-role** · SQS receive, S3 write, SSM read
- **scheduler-role** · send to exam-events

SI download straight from S3 via pre-signed URL (120 s)

email

A · Generate 20 copies (steps 1–7)

Controller clicks **Generate** → API saves 20 copies + fingerprints in **RDS** → 1 job per room → **SQS** → **CloudWatch** alarm: queue has work → **Auto Scaling** starts worker (0 → 1) → Worker renders PDFs → **S3** (via endpoint) → Progress in RDS → exam **READY** → Queue empty 5 min → worker stops

B · Release and download (steps 8–11)

Controller sets **Release at / Close at** → **EventBridge Scheduler** one-time schedules → At release: message → **exam-events** queue → API marks exam **RELEASED** → **SNS** release notice email → SI clicks Download (own centre, inside window) → API signs a 120 s **pre-signed URL** → Browser downloads PDF straight from **S3**

C · Leak trace

Leaked text pasted in **Trace Lab** → API scores all 20 copies in **RDS** → Names copy, room, seat, who downloaded → **SNS** leak-alert email

D · Failure handling

A generation job fails 3 times → Message moves to the **DLQ** → Alarm **dlq-not-empty** fires → **ops-alerts** email to the team

request / data flow · async / delivery · dashed red box = security group · public subnet (route to Internet Gateway) · private subnet (no internet route) · Numbers 1–12 match the flows A–D above